Flipkart – Big Billion Days: Week 1 Summary

IT457 Cloud Computing · Scalability & High-Traffic Events · Group 24

1. Case in one paragraph

Big Billion Days (BBD) is Flipkart's annual festival-season sale, running across September–October. It is India's equivalent of Black Friday/Cyber Monday and the single biggest stress test of Flipkart's platform. Compute demand swings between 2x and 10x of normal during sale events, so the core question is how to serve short, extreme spikes without paying for idle capacity all year.

2. Sources found

#	Source	Type	What it gives us
1	Google Cloud blog, "Reshaping Flipkart's technological landscape with a mammoth cloud migration"	Primary (vendor case study)	Data-platform migration to GCP in 8 months; 130+ billion messages/day; 60 TB/day normal vs 500 TB/day at peak; 85+ member migration team
2	Google Cloud blog, "How Flipkart gets the most out of every byte with Bigtable" (Aug 2024, Flipkart SDE III)	Primary (engineer-authored)	HBase → Bigtable migration for the fStream pipeline; auto-scaling; platform scaled 4x during BBD with no capacity or performance impact
3	Flipkart architect talk abstract, Hasgeek (Cloud Cost Optimization, 2023)	Primary (Flipkart architect)	Private cloud of millions of cores; 2x–10x demand swings; hybrid strategy to use public cloud for peaks
4	Rootconf 2015 talk abstract, Flipkart cloud platform architect	Primary (Flipkart)	Earlier in-house cloud designed for a million VMs and multi-Tbps traffic; spikes above 5x; avoiding hardware load balancers

#	Source	Type	What it gives us
5	Google Cloud–Flipkart partnership press release (8 Mar 2022)	Primary (press release)	Multi-year alliance for scaling during peak seasons, data analytics and ML
6	Aerospike customer story and Summit blog ("Journey to 90 million QPS")	Vendor case study	Aerospike run as a shared platform on Kubernetes (200+ clusters, sub-millisecond reads at peak); automated scaling via the Aerospike Kubernetes Operator
7	PingCAP (TiDB) case study	Vendor case study	Pain of sharded MySQL (downtime for resharding) and move to a self-scaling distributed SQL database
8	DEV Community HLD post; Substack and Hashnode summaries	Secondary (blogs)	Useful for architecture-diagram ideas (microservices, Kubernetes autoscaling, Kafka). Treat as unverified
9	DQ India, "How Flipkart handles Big Billion Days traffic" (8 Oct 2026)	Secondary (trade press)	Synthesis of Aerospike, Google Cloud and Flipkart accounts: separate layers (Kubernetes/AKO, Aerospike, fStream), tail latency, chaos testing, payments at ~1,000x peak load; clarifies the 95M vs 90M figures

Caveats: the two Aerospike figures are different measurements, not a contradiction: 95 million database transactions/sec (Aerospike report, Nov 2024) versus 90 million queries/sec across 50+ use cases and three data centres (Nov 2025). We quote both with their dates and never compare them year on year. One student document online claims Flipkart moved to a hybrid cloud on AWS, which conflicts with the Google Cloud sources, so we do not use it. Rows 8 and 9 are for ideas and context only; hard figures come from the primary sources. No source confirms Flipkart's 2026 configuration, so all figures are historical.

3. Salient points of the case

- **Problem:** very sharp, predictable but extreme peaks (2x–10x compute); idle private capacity outside sale periods is costly; sharded MySQL and manual cluster operations were hard to scale safely at peak.
- **Architecture:** microservices on Kubernetes with autoscaling; event streaming (Kafka, internal fStream on Spark and Flink); NoSQL and distributed stores (Aerospike, Bigtable/HBase, TiDB) for high-throughput data paths. Layers are separated: Kubernetes and the Aerospike Kubernetes Operator orchestrate clusters, Aerospike serves low-latency reads and writes, and workloads are isolated so one spike does not degrade another.
- **Hybrid cloud:** baseline load on Flipkart's own private cloud, bursting to public cloud (Google Cloud) for peak events, trading higher unit cost for elasticity.
- **Data scale:** 130+ billion messages/day and 500 TB/day of data at peak versus 60 TB/day normally.
- **Reliability:** continuous chaos testing (killing pods, simulating partial failures) and a focus on tail latency (the slowest requests) rather than averages.
- **External dependencies:** payments reportedly see nearly 1,000x normal request volume at peak (Flipkart payments VP, Business Standard interview, Sept 2026, reported secondhand), so banks and gateways must scale too.
- **Outcomes:** 4x scale-up during BBD with no capacity concerns; 95 million database transactions/sec and 200+ Aerospike clusters with sub-millisecond reads at peak; zero-downtime rolling upgrades, with deployments cut from days to minutes; data platform migration finished in 8 months.